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VENDOR SPECIFICATION MEMO: SCADA EMERGENCY BURST PROTOCOL (REV 2.4)

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[PHYSICAL LAYER]

- Modulation: Continuous-phase dual-tone modulation with oscillator drift:

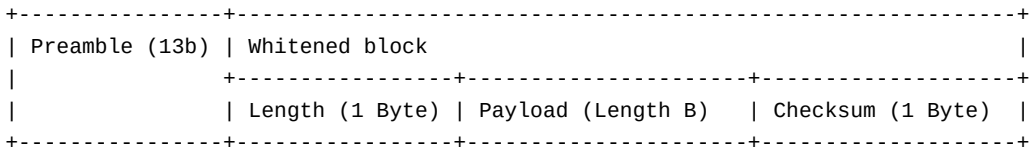
$$f_0(t) = f_{\text{base}} + \Delta f_{\text{drift}} \sin(\omega_d\, t)$$

$$f_1(t) = f_0(t) + \Delta f_{\text{shift}}$$

  Here t is elapsed time in seconds from the start of the transmitted burst.
  Carrier and drift parameters are omitted from this partial memo.

- Transmit Feed: Differential balanced line (opposite polarity across Channels A/B).
- Symbol Rate: Baseband RC timer calibrated at approximately 37.3 Baud.

[DATA LINK LAYER / PACKET FRAMING]



- Frame Preamble: 13-bit sequence.
- Data Whitener: Additive 7-stage LFSR; applied to the length, payload, and checksum as one continuous MSB-first bitstream. See recovered exchange below.
- Integrity Check: 8-bit CRC (SMBus standard generator polynomial 0x07).

[RECOVERED ENGINEERING EXCHANGE]

Engineer A: "That ’proprietary’ mask? x to the seventh, x to the fourth, and the lonely one. Taps seven and four choose the next mask bit."

Engineer B: "What state does it wake in?"

Engineer A: "Every one of the seven lamps is lit. Start at the most significant end of each byte; the length goes first and the check byte last."

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